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# Quick Start

The example code provided in this quick start guide is for educational and demonstration purposes only. It may not represent best practices for production use.

## Breaking Changes Notice

If you've just updated the package, it is recommended to check the [changelogs](#) for information on breaking changes.

## Prerequisites

- Unity Version: Unity 6.0+ (tested on Unity 6.3).
- Importer Dependencies (Install at least one):
  - [glTFast](#) for GLTF/GLB support.
  - [UniGLTF](#) for alternative GLTF/GLB support.
  - [UniVRM](#) for VRM support.
- Other Optional Dependencies:
  - Unity's Animation module for animator post-processing.

## Key Concepts

- **Data Loaders:** Fetch raw avatar data (model, metadata, renders) from files or URIs.
- **Importers:** Convert raw data into Unity GameObjects.
- **Post-Processors:** Customize loaded avatars (e.g., add animators, override materials).
- **AvFlow:** Chains loaders, importers, and post-processors for a complete loading pipeline.

Avatars require a specific structure:

- Model file (e.g., `model.glb`, `model.vrm`).
- `metadata.json` with ID, type, gender, etc.
- Optional renders: `full.jpg/png`, `bust.jpg/png`.

## Basic Usage

Use `AvFlow` to build a loading pipeline.

## Example: Load from Local File

```
public class AvatarLoader : MonoBehaviour
{
    async void Start()
    {
        // FileAvDataLoader supports auto-detection of the model format by file extension
    }
}
```

```

var flow = new AvFlow.Builder()
    .From(new FileAvDataLoader("path/to/avatar/dir")) // Directory with model,
metadata, renders
    .Using(new GLTFastAvImporter()) // Or UniGLTFAvImporter, UniVRM10AvImporter
    .Build();

LoadedAv? avatar = await flow.TryRunAsync(); // Or `RunAsync()` for explicit
throwing errors.
if (avatar != null)
{
    avatar.GameObject.transform.position = Vector3.zero; // Position in scene
    // Use avatar.GameObject, avatar.Metadata, etc.
}
else
{
    Debug.LogError("Failed to load avatar");
}
}
}

```

## Example: Load from URI (Remote)

```

var flow = new AvFlow.Builder()
    .From(new URIAvDataLoader(
        "https://example.com/model.glb",
        "https://example.com/metadata.json",
        AvModelFileExtension.GLB // Model format must be explicitly declared
    ))
    .Using(new GLTFastAvImporter())
    .Build();

LoadedAv avatar = await flow.RunAsync(); // Throws on failure

```

## Adding Post-Processors

Post-processors run after import. Synchronous ones run first, then async.

```

// In AvFlow.Builder
.ProcessWith(new AnimatorPostProcessor() { AnimatorController = myController }) //
Add animator
.ProcessWithAsync(new CachePostProcessor("cache/dir")) // Cache downloaded model to
local dir

```

- **Material Overrides/Shader:** Use `MaterialOverridePostProcessor` or `ShaderSwapPostProcessor` with configs (create via ScriptableObject).
- **Conditional:** Wrap with `ConditionalPostProcessor` for logic-based processing.

Sync processors (e.g., `AnimatorPostProcessor`) always run before async ones (e.g., `CachePostProcessor`), regardless of addition order.

The `AvPost` class has static methods for fluent-style creation of post-processors.

## Multi-Importer Example with More Post-Processors

```
[SerializeField] RuntimeAnimatorController _animator;
[SerializeField] Avatar _femAvatar;
[SerializeField] Avatar _mascAvatar;
[SerializeField] bool shouldLoadGltf;

new AvFlow.Builder()
.From(
    new FileAvDataLoader(
        shouldLoadGltf
        ? Path.Join(Application.streamingAssetsPath, "GLTFModelFolder")
        : Path.Join(Application.streamingAssetsPath, "VRMModelFolder")
    )
).Using(
    new UniVRM10AvImporter(),
    new GLTFFastAvImporter(),
    new UniGLTFAvImporter()
    {
        ImporterContextSettings = new UniGLTF.ImporterContextSettings(invertAxis:
UniGLTF.Axes.X) // For compatability with glTFast
    }
).ProcessWith(
    AvPost.ConfigureAnimator(_animator, _mascAvatar, new Dictionary<AvGender, Avatar>()
    {
        { AvGender.Feminine, _femAvatar },
        { AvGender.Masculine, _mascAvatar },
    })),

    AvPost.WhenLoadedWith<FileAvDataLoader>(
        AvPost.Do(static (_, _) => print("LOADED AVATAR FROM FILES"))
    ),
    AvPost.WhenLoadedWith<URIAvDataLoader>(
        AvPost.Do(static (_, _) => print("LOADED AVATAR FROM URIS"))
    ),

    AvPost.WhenImportedWith<UniVRM10AvImporter>(
```

```

        AvPost.Do(static (_, _) => print("LOADED VRM AVATAR"))
    ),
    AvPost.WhenImportedWith<UniGLTFImporter>(
        AvPost.Do(static (_, _) => print("LOADED UNIGLTF AVATAR"))
    ),
    AvPost.WhenImportedWith<GLTFastImporter>(
        AvPost.Do(static (_, _) => print("LOADED GLTFAST AVATAR"))
    )
).ProcessWithAsync(new CachePostProcessor(Application.temporaryCachePath))
.Build();

```

## Handling Renders and Metadata

- Access `avatar.FullRender` and `avatar.BustRender` for thumbnails.
- `avatar.Metadata` includes ID, gender, type (e.g., `AvType.HumanoidFullBody`).

## Cleanup

Always dispose loaded avatars:

```

avatar.Dispose(); // Destroys GameObject, renders, post-processor created materials, etc.

```

For full API docs, please see <https://uralstech.github.io/AvLoader/api/Uralstech.AvLoader.html> or [APIReferenceManual.pdf](#) in the package documentation for the reference manual.

You can implement `IAvDataLoader` or `IAvImporter` for custom sources/importers (e.g., cloud APIs, FBX import).

## Advanced - Capabilities

Capabilities are interfaces in the `Uralstech.AvLoader.Capabilities` namespace for `LoadedAv` that expose functionality guaranteed to exist because a the avatar importer created it, independent of later configuration or user modifications.

They act as a reliable contract so consuming code can safely detect and use importer-provided features, such as VRM LookAt, VRM Expressions, or an importer-generated Animator, without guessing or inspecting arbitrary components. Capabilities describe availability and guarantees only, not ownership, configuration, or policy, and they are implemented exclusively by importer-backed `LoadedAv` objects, never by post-processors or user code.

**Capabilities should *not* be accessed by using interface methods directly, but by casting the `LoadedAv` object to the interface.** You can also use the utility methods defined in `LoadedAv` for doing so:

```
LoadedAv avatar = ...  
if (!avatar.TryGetCapability(out IBlendShapeProvider? provider))  
    return;  
  
if (provider.HasWeight("blink"))  
    provider.SetWeight("blink", 1f); // Close the avatar's eyes.
```